

Course title	Programming and Data Structures including Lab	Course No.							
Department		Structure	3	0	0	3	9	15	
Offered for		Status	Core √		Elective				
Faculty		Type	New √		Modification				
Prerequisite e									

Objectives: To introduce students to basic data structures and algorithms

Course contents:

1. Introduction to algorithms and data structures
2. Basic data structures: stacks, linked lists, queues
3. Basic algorithms: Simple sorting
4. Advanced sorting algorithms
5. Binary Trees, Hash tables, Heaps
6. Dynamic Programming algorithms
7. Greedy algorithms
8. Introduction to graphs
9. Graph algorithms: BFS, DFS, A*
10. Advanced topics: Parallel programming, Randomized algorithms

Textbooks:

Lafore, R., Broder, A., & Canning, J. (2022). Data Structures & Algorithms in Python. Addison Wesley.

Reference books:

Skiena Steven, S. (2008). The Algorithm Design Manual. Springer.